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Climate Smart Agriculture: A False Solution to address Climate Crisis

Climate Smart Agriculture is a concept developed by the Food and Agriculture Organization (FAO) in 2010, with intentions

to solve problems caused by Climate Change, Global Agriculture and its stake holders on each other (FAO, 2010).

Climate Smart Agriculture is defined as an approach to agriculture which aims to sustainably increase agriculture productivity and income, adapt and build resilience to climate change, reduce and/or remove greenhouse gas emissions and enhance achievement of national food security and development goals; where productivity, adaptation and mitigation are the foundational pillars to achieve this goal. (Lipper et al., 2014).

The definition of Climate Smart Agriculture is ambiguous when it comes to implementation in the ground level. There is a lack of consensus about what practices and technologies should be considered in Climate Smart Agriculture. As the definition of Climate Smart Agriculture points out, one of its goals is to increase productivity and promote food security; hence some argue that any agriculture practices that improves productivity and efficient use of resources can be considered climate smart. (Neufeldt et al., 2013) Nonetheless, such an approach overlooks the environmental impact created in the process. On the other hand, the relationship between conservational approach and Climate Smart Agriculture is overlooked. Organizations that formulate policies at the top level do not give enough attention to indigenous agroecological methods. Many of the conservation techniques such as minimum tillage, different method of crop establishment, nutrition and irrigation management etc. can improve crop yield and reduce GHG at the same time with little to no additional cost. (Branca et al., 2011; Jat et al., 2014; Sapkota et al., 2015). Such inconsistencies in the understanding of Climate Smart Agriculture is prevalent globally and has created mechanisms for exploitation of small scale farmers and our environment as a whole.

This top to bottom approach i.e. implementation of policies recommended by large international organizations to millions of small-scale farmers is not a new phenomenon. Such policies which are backed by large amounts of funding show a lot of promise on paper. However, previous examples of such mechanisms to improve the global agricultural sector like the Green revolution of the 1960's, similar to Climate Smart Agriculture in its framework, had many severely damaging consequences to wide groups of people. (Sherpa, 2017) The major victims of these consequences were small holder farmers from developing countries and various marginalized minority groups.

Climate Smart Agriculture is a marketing scheme presented as a solution to governments by corporations to deal with climate related issues in agriculture but in fact is a rebranding of green revolution practices which have so far heavily contributed to climate change. (Sherpa, 2017)

Thus, implementation of initiatives such as Climate Smart Agriculture should be done with rigorous and extensive analysis as it might have horrific costs to all the stakeholders involved in the long run.

Globally, the current model of Climate Smart Agriculture advocates for agricultural mechanisms which mainly include but are not limited to large scale mono-cropping, hi-technological investments and large chemical inputs to the food production system which require tremendous amounts of capital and centralized control. (Sherpa, 2017) This approach is bound to create injustice to the small holder and family farmers practicing subsistence agriculture, who comprise about 80% of Asian and Sub-Saharan African farmers (FAO factsheet), as they do not have the capability to invest such

remarkable amounts. These obstacles will cause large number of small scale farmers to become disenfranchised from their traditional occupation and will most likely cause them to suffer from forced migration. Simultaneously, this creates a perfect opportunity for large agri-businesses to take over. In fact, Global Alliance for Climate Smart Agriculture (GAClimate Smart Agriculture) was created under FAO to provide a platform for government and other organizations to form “transformational partnerships” by sharing information, knowledge and expertise to achieve the goals of Climate Smart Agriculture. Paradoxically, its members are mainly big chemical fertilizer companies and agribusinesses that are far from being advocates of climate change or sustainability, but instead are some of the worst polluting multinational companies in the world. (Sherpa, 2017) Moreover, such forums lack the inclusion and representation of farmers from small backgrounds and developing countries which questions its legitimacy on combating climate change for people of all groups and backgrounds. Ironically, Global Alliance for Climate Smart Agriculture have put corporations with horrible tracks records, when it comes to fighting Climate Change, on the center stage for developing, sharing and executing policies to combat Climate Change. This begs us stop and ask “If Climate Smart Agriculture is another tool for corporations and agri-businesses to keep profiting at the cost of the environment and marginalized groups across the globe?”

Increasingly, many Civil Society Organizations (CSOs) have expressed concerns that the term “Climate Smart Agriculture” is designed to green-wash agricultural practices that have the propensity to harm future food production. (Anderson, 2014)

To demonstrate resilience against Climate Smart Agriculture, a group of more than 100 civil society organizations have signed a petition rejecting GACAS pointing out that industrial agriculture approaches which increase GHG emissions are now permitted to use the climate-smart label to promote their agri-businesses as solutions to climate change. (Climate Smart Agriculture Concerns)

The world largest peasant farmers’ movement, La Vie Campesina, with many other farmer organizations, stood together in solidarity to reject Climate Smart Agriculture as an effort to put agribusiness agenda under a green mask. (Anderson, 2014).



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CLIMATE SMART AGRICULTURE IN NEPAL

In the context of Nepal, recently there has been a realization for the need of planned efforts to address the challenges of climate change and its impacts to the people of Nepal and the environment. In light of such realization the Government of Nepal (GoN) has enacted a National climate change Policy 2076, implemented National Adaptation program of Action (NAPA), initiated a Pilot Program on Climate Research (PPCR). Similarly, to tackle the problems in the Agriculture sector the Ministry of Agriculture Development (MoAD) has released Agriculture Development Strategy 2015 (ADS). (Khatri-Chhetri et al., 2017) There is almost no mention of Climate Smart Agriculture explicitly in such official documents. ADS mentions Climate Smart Agriculture briefly as an approach to practice climate resilient agriculture in order to improve productivity and food security without any explanation on how Climate Smart Agriculture would be executed in the context of Nepal. At present, Climate Smart Agriculture has been trending globally as “the technique” that can be implemented as a solution for the effect of climate change on agriculture and food security. However, as we have already explored there are countless agriculture practices that could qualify under the principle of Climate Smart Agriculture.

The concerned government authorities need to explicitly clarify what sort of approach is Nepal going to follow under the practice of Climate Smart Agriculture.

There are no meaningful criteria for what can and cannot be called “climate-smart”. The looseness of the term permits anyone to use this term to describe their activity with no regards to the impact it creates socially or environmentally (Anderson, 2014). Such ambiguity regarding the application mechanism of Climate Smart Agriculture presents an opportunity for powerful agro-corporations to swiftly lobby for the use of extensive chemical inputs and risky high-tech methods under the banner of Climate Smart Agriculture. Government of Nepal needs to be aware of such prospects and clearly define what constitutes as Climate Smart Agriculture.

An assessment conducted by Chhetri, Poudel and Shirsath on Climate Smart Agriculture options in Nepal states that Climate Smart Agriculture policies in Nepal primarily focus on implementation of better agricultural practices and technologies, livelihood diversification and capacity building. However, reduction of GHG emission is not a top priority (Khatri-Chhetri et al., 2017). When a survey of locals was taken to determine the priority given to the different pillars of Climate Smart Agriculture, the mitigation objective was the least prioritized in comparison to food security and adaptation. The current scenario of policy efforts in Nepal seems to only focus on increasing agricultural productivity without placing any importance on mitigation measures (Khatri-Chhetri et al., 2017). Again, such imbalanced approach is a perfect invitation for agri-business which have the capacity to spend high amounts of capital on high-tech and chemical inputs which will improve crop yield in the

short-run. Nevertheless, introduction of agri-businesses will also create more environmental problems and social division because of their profit maximization approach.

The farming systems and farmtopology in Nepal are very diverse and the involvement of women and marginalized groups in Agriculture is soaring. This requires location specific climate smart interventions that includes issues of gender and social inclusion (Khatri-Chhetri et al., 2017). Having understood that the solution, in the case of Nepal, is geographically contextual; a narrow techno centric approach with high use of external input in the disguise of Climate Smart Agriculture is the false solution.

This will create dependency on so called new technologies through their complete packages that include prescription of climate smart varieties of seeds, inputs and credit while ignoring traditional and true adaptive farming techniques and stewardship of indigenous seed varieties in practice by farmer specific to that particular geography (Thapa, 2014).

Contrary to what the government and agribusinesses have made us believe, people from local level have presented real solutions in the form of local energy sovereignty, food security via agro-ecology, drastic change in production and consumption patterns and regaining control over resources which include land, forest, water etc. (Sherpa, 2017). Such holistic approach will lead us to food security whilst fulfilling our sustainability goals and help us adapt to and mitigate climate change caused by agriculture practiced in our country.

In terms of the sustainability claims both Climate Smart Agriculture and agroecology appear to be interchangeable concepts. Similar to agroecology, Climate Smart Agriculture promotes sustainable increase in agricultural productivity and income and acknowledges that it is not a practice that can be universally applied but rather an approach which involve different measure based on local context. This statement makes an attempt to prove the existence of an overlap between Climate Smart Agriculture and agroecology. Nevertheless, this is far from reality (Sherpa, 2017). Firstly, under Climate Smart Agriculture there are no climate, environmental or social criteria that needs to be fulfilled for a project to be implemented (Anderson, 2014). Secondly, Climate Smart Agriculture endorses use of proven harmful technologies which is contrary to agroecology like chemical herbicides and GMOs. They focus on carbon offsetting schemes which allow fraudulent credits to be earned from Climate Smart Agriculture projects to offset emissions and allow polluters to continue polluting. (Sherpa, 2017).

It appears that Climate Smart Agriculture is designed to expand carbon markets and serve the interest of agribusinesses and the financial industry. Hence, most of the agricultural practices

promoted under Climate Smart Agriculture is reliant on corporate technology, use of chemical fertilizers and use of high external input which ultimately benefits few large holder farmers and serve the interest of big agribusiness.

For developing countries like Nepal this is a big conundrum. On one hand, initiatives like Climate Smart Agriculture backed by renowned international organizations promises a better future through food security, adaptation and resilience to climate change and mitigation of its effects on the long run. On the other hand, awareness of the hidden agendas curbed under different international programs promoted by renowned International Organizations. with its Climate Smart Agriculture, Carbon credit trading schemes like REDD and REDD plus etc. make you question the authenticity of such initiatives.

Developing countries around the world need open and free access to technology that can be adapted, improved and used by small farmers without threats of patents and intellectual property hanging over their heads (Thapa, 2014). They need the support of international community to collaborate and develop methods to conduct agriculture which is in synergy with our environment that truly promotes sustainability. However, this is not what we are seeing today. Vested interest of private corporations and big agri-businesses serve to commodify and privatize the functions of nature and destroy the ecosystem. Their initiatives will not reduce emissions nor address social crisis due to climate change but rather allow business as usual and create corporate profits in the name of combating climate change (Sherpa, 2017). Today businesses whose activities result in dire social impacts on farmers and communities such as those driving land grabbing or promoting Genetically Modified seeds are the once who have claimed the title of 'climate smart'.

It is high time we reclaim the meaning of practicing Climate Smart Agriculture. In fact, we need to ask ourselves if we really need Climate Smart Agriculture in our vocabulary. There is plenty of evidence that such "initiatives" endorsed by international organizations are yet another strategy by powerful agribusiness corporation to prop up industrial agriculture just like the Green revolution through green-washing.

Alternatively, climate-resilient sustainable agriculture approaches i.e. fundamental practices of agroecology is needed to help food systems adapt to and mitigate climate change (Anderson, 2014). Agroecology has already been recognized as the solution to address the issue of agriculture and climate change by many Civil Society Organizations as well as numerous farmer groups. There is enough research data and evidence which indicate that peasant based agroecological approaches have immense mitigation and adaptation potential and can make significant contribution towards reduction of GHG emissions. It uses less fossil fuel based inputs and has a better carbon footprint than green agriculture based practices which heavily depend upon energy intensive fertilizer, chemical and concentrated feed. Agroecology therefore leads to better carbon sequestration in the soil and can contribute in mitigating climate related risks (Thapa, 2014).

Nepal and countries of similar profile need to be aware of the consequences that come with adopting Climate Smart Agriculture. Growing interests in Climate Smart Agriculture globally is creating a confusion in the food and climate movement. Climate Smart Agriculture's clever name should not distract us from its inherent power imbalances and that it may create more risks than benefits to the climate and our food systems.

Just because a policy is validated by internationally acclaimed organizations does not mean it is the right approach.

Additionally, we need to stop being naïve and critically investigate who benefits from such initiatives and which groups are oppressed in the process. However, the fight against Climate change and its devastating effects is our ultimate adversary and it must not stop. Agroecological approaches that are proven to improve farm yields, livelihoods and environment of small scale farmers while adapting to and mitigating effects of Climate change based on local needs is a better substitute to adopting climate smart agriculture.

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